

SHARP RECIPROCAL-MOMENT INEQUALITIES FOR CRITICAL POINTS OF POLYNOMIALS WITH COLLINEAR ZEROS

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ABSTRACT. Let p be a complex polynomial of degree $n \geq 2$ whose zeros lie on one affine line, let a be a zero of p , and let $w_1, \dots, w_{n-1}$ be the critical points of p , counted with multiplicity. If $D > 0$ is the diameter of the zero set, then for every real number $\lambda \geq 1$,

$$\sum_{j=1}^{n-1} \frac{1}{|a - w_j|^\lambda} \geq (n-1) \left(\frac{2}{D} \right)^\lambda,$$

where a term with $w_j = a$ is interpreted as $+\infty$. The constant is sharp. For $\lambda = 1$, equality holds precisely when a is one endpoint of the zero interval and all the remaining zeros coincide at the other endpoint; for $\lambda > 1$, equality holds precisely in degree two. Since a collinear zero set contained in the closed unit disk has diameter at most two, the theorem proves the Tang–Zhang reciprocal-moment strengthening of Sendov’s conjecture for every polynomial with collinear zeros. The proof is elementary: weak Rolle interlacing separates the positive and negative parts of the logarithmic-derivative identity at the distinguished zero, after which a one-dimensional diameter estimate and the power-mean inequality complete the argument.

Generative-AI disclosure. This manuscript was prepared by guiding ChatGPT 5.6 Sol Pro Chat. The article, including its mathematical exposition and bibliography, was completed with AI assistance. The author made many modifications to the generated manuscript and assumes responsibility for the final version.

1. INTRODUCTION

The geometric relation between the zeros of a polynomial and the zeros of its derivative is one of the classical themes of one-variable complex analysis. The Gauss–Lucas theorem gives the fundamental convex-hull inclusion, while Rolle’s theorem gives the much more rigid alternation of zeros and critical points in the real case. Standard accounts may be found in Marden’s monograph [Mar66, Chapter II] and in Rahman and Schmeisser [RS02]; broader surveys of the Sendov problem are given by Marden [Mar83] and Rahman and Schmeisser [RS02]. Related global comparison principles include Schoenberg’s conjectural analogue of Rolle’s theorem [Sch86], Schmeisser’s majorization theorem [Sch04], and Totik’s survey of the geometry and distribution of critical points [Tot21].

Sendov’s conjecture asserts that if all zeros of a polynomial lie in the closed unit disk, then every zero has a critical point within distance one. Early special cases and structural estimates were obtained by Brannan [Bra68], Rubinstein [Rub68], Meir and Sharma [MS69], Schmeisser [Sch69, Sch71], Saff and Twomey [ST71], and Phelps and Rodriguez [PR72]. Quantitative, variational, and degree-dependent developments include the work of Bojanov, Rahman, and Szynal [BRS85], Miller [Mil90, Mil08], Borcea [Bor98], Brown and Xiang [BX99], Dégot [Deg14], Chalebgwa [Cha20], Sofi, Ahanger, and Gardner [SAG20], Tao [Tao22], and

2020 *Mathematics Subject Classification.* Primary 30C10; Secondary 26C10, 30A10.

Key words and phrases. Sendov’s conjecture; critical points of polynomials; collinear zeros; Rolle interlacing; reciprocal moments; sharp inequalities.

Zhang [Zha26]. A recent preprint of Mazur presents a computer-assisted proof based on reciprocal critical-point coordinates, an antiderivative polynomial, finite-product defect estimates, and exact scalar certificates [Maz26].

The collinear-zero case has a particularly transparent history. Phelps and Rodriguez proved the ordinary Sendov conclusion for polynomials with real zeros [PR72, Theorem 6]. Sofi and Shah later treated zeros lying on a line or a circle [SS23]. Miller obtained the quantitative refinement

$$d(p, a) \leq 1 - \frac{1}{2}|a|(1 - |a|)$$

for the line case, in the usual normalization of the unit disk [Mil25, Theorem 7]. The surrounding real-polynomial geometry is also connected with the work of Brown and Powell [BP07].

More recently, Tang and Zhang proposed a moment-theoretic strengthening of Sendov's conjecture. Their formulation is local at a prescribed zero and belongs naturally to the family of negative-order Schoenberg-type inequalities considered in their work [TZ25, Conjecture 1.10].

Conjecture 1.1. *Let p be a polynomial of degree $n \geq 2$ whose zeros lie in the closed unit disk, let a be a zero of p , and let $w_1, \dots, w_{n-1}$ be the zeros of p' , counted with multiplicity. Then, for every $\lambda \geq 1$,*

$$\sum_{j=1}^{n-1} \frac{1}{|a - w_j|^\lambda} \geq n - 1.$$

The purpose of the present paper is to establish Conjecture 1.1 completely for polynomials whose zeros are collinear. In fact, the one-dimensional geometry yields a stronger, scale-invariant inequality in which the unit-disk normalization disappears altogether. The natural geometric quantity is the diameter of the zero set.

Theorem 1.2. *Let p be a polynomial of degree $n \geq 2$ whose zeros are contained in a line in the complex plane, and suppose that the diameter D of its zero set is positive. Let a be any zero of p , and let $w_1, \dots, w_{n-1}$ denote the zeros of p' , counted with multiplicity. Then, for every $\lambda \geq 1$,*

$$\sum_{j=1}^{n-1} \frac{1}{|a - w_j|^\lambda} \geq (n - 1) \left(\frac{2}{D} \right)^\lambda.$$

The constant is sharp.

Since a subset of the closed unit disk has diameter at most 2, Theorem 4.1 immediately implies Conjecture 1.1 for every polynomial with collinear zeros. The stronger dependence on D also shows that the unit-disk estimate is not the intrinsic one-dimensional statement: the relevant scale is the diameter of the zero set rather than the radius of an ambient disk.

We shall also determine all equality cases in Theorem 4.1. No distinctness assumption is imposed on the zeros of p or on the critical points, and all zeros and critical points are counted with their algebraic multiplicities throughout.

The proof of Theorem 4.1 has two ingredients. First, Rolle interlacing and the logarithmic derivative yield a signed reciprocal-balance inequality at the distinguished zero. Second, the number of zeros on the two sides of a forces one of the two signed reciprocal masses to be at least $(n - 1)/D$. The power-mean inequality then gives every exponent $\lambda \geq 1$. The argument also clarifies the role of the antiderivative coordinates used by Mazur: in the collinear case, their complex phases reduce to signs, and interlacing supplies exactly the sign separation needed to pass from an algebraic identity to an absolute-value estimate.

2. INTERLACING ON A LINE

A finite subset of $\mathbb{C}$ is called collinear if it is contained in an affine line. Rigid motions of the plane preserve zero–critical-point distances and commute with differentiation up to a nonzero scalar. Consequently, whenever the zeros of p are collinear, we may rotate and translate the common line to the real axis and then multiply the polynomial by a nonzero constant. We shall use this reduction without further comment.

The following weak form of Rolle interlacing includes all multiplicities and is the only location theorem required below.

Lemma 2.1. *Let p be a real polynomial of degree $n \geq 2$. List its zeros with multiplicity as*

$$x_1 \leq x_2 \leq \cdots \leq x_n.$$

Then the zeros of p' can be listed with multiplicity as

$$y_1 \leq y_2 \leq \cdots \leq y_{n-1}$$

so that

$$x_i \leq y_i \leq x_{i+1}, \quad 1 \leq i \leq n-1.$$

If $x_i < x_{i+1}$, at least one critical point lies in the open interval (x_i, x_{i+1}) .

Proof. Let $\xi_1 < \cdots < \xi_s$ be the distinct zeros of p , with multiplicities $\mu_1, \dots, \mu_s$. The derivative has a zero of multiplicity $\mu_r - 1$ at ξ_r , and Rolle's theorem supplies at least one zero of p' in every open interval (ξ_r, ξ_{r+1}) . These account for

$$\sum_{r=1}^s (\mu_r - 1) + (s - 1) = n - 1$$

zeros, which is the full degree of p' . Ordering these zeros gives the asserted weak interlacing. The final statement is the ordinary strict form of Rolle's theorem on an interval with distinct endpoints. $\square$

The interlacing lemma is a location statement. To obtain the reciprocal-moment estimate, we next combine it with an exact algebraic identity at a prescribed simple zero.

3. SIGNED RECIPROCAL BALANCE

Let $a = x_k$ be a simple zero of the real polynomial p . With the notation of Lemma 2.1, put

$$P_- := \sum_{i < k} \frac{1}{a - x_i}, \quad P_+ := \sum_{i > k} \frac{1}{x_i - a},$$

and

$$Q_- := \sum_{i < k} \frac{1}{a - y_i}, \quad Q_+ := \sum_{i \geq k} \frac{1}{y_i - a}.$$

Empty sums are understood to be zero. Since a is simple, none of the denominators involving y_i vanishes.

Proposition 3.1. *In the preceding notation,*

$$Q_- \geq P_-, \quad Q_+ \geq P_+, \quad Q_- - Q_+ = 2(P_- - P_+). \quad (3.1)$$

Consequently,

$$\sum_{j=1}^{n-1} \frac{1}{|a - y_j|} = Q_- + Q_+ \geq 2 \max\{P_-, P_+\}. \quad (3.2)$$

Proof. For $i < k$, interlacing gives $x_i \leq y_i < a$, hence

$$\frac{1}{a - y_i} \geq \frac{1}{a - x_i}.$$

For $i \geq k$, it gives $a < y_i \leq x_{i+1}$, hence

$$\frac{1}{y_i - a} \geq \frac{1}{x_{i+1} - a}.$$

Summation proves the first two inequalities in (3.1).

Write $p(z) = (z - a)g(z)$. Since $g(a) \neq 0$,

$$\frac{p''(a)}{p'(a)} = 2 \frac{g'(a)}{g(a)} = 2 \sum_{i \neq k} \frac{1}{a - x_i} = 2(P_- - P_+).$$

On the other hand, factoring p' at its zeros gives

$$\frac{p''(a)}{p'(a)} = \sum_{j=1}^{n-1} \frac{1}{a - y_j} = Q_- - Q_+.$$

This proves the identity in (3.1).

If $P_- \geq P_+$, then

$$Q_- + Q_+ = 2Q_+ + 2(P_- - P_+) \geq 2P_-.$$

If $P_+ \geq P_-$, the symmetric calculation gives $Q_- + Q_+ \geq 2P_+$. Thus (3.2) follows. $\square$

The preceding proposition separates the proof into an algebraic signed-balance identity and a one-dimensional geometric estimate. The latter can be stated in a count-sensitive form before the diameter is introduced.

Corollary 3.2. *Let $\alpha = x_1$, $\beta = x_n$, and suppose that $a = x_k$ is simple. Put $\ell = k - 1$ and $r = n - k$. Then*

$$\sum_{j=1}^{n-1} \frac{1}{|a - y_j|} \geq 2 \max \left\{ \frac{\ell}{a - \alpha}, \frac{r}{\beta - a} \right\},$$

where a quotient with zero numerator is omitted.

Proof. Every zero to the left of a is at distance at most $a - \alpha$, and every zero to the right is at distance at most $\beta - a$. Hence

$$P_- \geq \frac{\ell}{a - \alpha}, \quad P_+ \geq \frac{r}{\beta - a}.$$

The conclusion follows from Proposition 3.1. $\square$

4. THE SHARP DIAMETER INEQUALITY

We now return to complex polynomials with collinear zeros. For a finite multiset $Z \subset \mathbb{C}$, its diameter is

$$\text{diam } Z := \max\{|z - \zeta| : z, \zeta \in Z\}.$$

Multiplicity does not affect the diameter.

Theorem 4.1. *Let p be a complex polynomial of degree $n \geq 2$ whose zeros are collinear, and suppose that its zero set has diameter $D > 0$. Let a be a zero of p , and let $w_1, \dots, w_{n-1}$ be the zeros of p' , counted with multiplicity. Then, for every real number $\lambda \geq 1$,*

$$\sum_{j=1}^{n-1} \frac{1}{|a - w_j|^\lambda} \geq (n-1) \left(\frac{2}{D} \right)^\lambda. \quad (4.1)$$

If $a = w_j$ for some j , the corresponding term is interpreted as $+\infty$.

Proof. If a is a multiple zero of p , then $p'(a) = 0$ and the left-hand side of (4.1) is infinite. We may therefore assume that a is simple.

Apply a rigid motion sending the common zero line to $\mathbb{R}$. Order the zeros as in Lemma 2.1, with $a = x_k$, and write

$$\alpha = x_1, \quad \beta = x_n, \quad D = \beta - \alpha, \quad \ell = k - 1, \quad r = n - k.$$

If $a = \alpha$, then $r = n - 1$ and Corollary 3.2 gives

$$\sum_{j=1}^{n-1} \frac{1}{|a - y_j|} \geq \frac{2(n-1)}{D}.$$

The case $a = \beta$ is identical.

Suppose now that $\alpha < a < \beta$. Since $\ell + r = n - 1$ and

$$(a - \alpha) + (\beta - a) = D,$$

at least one of the two inequalities

$$\frac{\ell}{a - \alpha} \geq \frac{n-1}{D}, \quad \frac{r}{\beta - a} \geq \frac{n-1}{D}$$

must hold. Indeed, if both failed, adding the resulting strict inequalities would give $\ell + r < n - 1$. Corollary 3.2 therefore yields

$$\sum_{j=1}^{n-1} \frac{1}{|a - w_j|} \geq \frac{2(n-1)}{D}. \quad (4.2)$$

Set $m = n - 1$ and $t_j = |a - w_j|^{-1}$. For $\lambda \geq 1$, the power-mean inequality and (4.2) give

$$\sum_{j=1}^m t_j^\lambda \geq m^{1-\lambda} \left(\sum_{j=1}^m t_j \right)^\lambda \geq m \left(\frac{2}{D} \right)^\lambda.$$

This is (4.1). □

The scale-invariant theorem immediately implies the reciprocal-moment conjecture in the unit disk.

Corollary 4.2. *Let p be a complex polynomial of degree $n \geq 2$ whose zeros are collinear and lie in the closed unit disk. For every zero a of p , every $\lambda \geq 1$, and the critical points $w_1, \dots, w_{n-1}$ of p ,*

$$\sum_{j=1}^{n-1} \frac{1}{|a - w_j|^\lambda} \geq n - 1. \quad (4.3)$$

Thus the Tang–Zhang conjecture holds for every polynomial with collinear zeros.

Proof. The diameter D of a subset of the closed unit disk satisfies $D \leq 2$. Therefore Theorem 4.1 gives

$$\sum_{j=1}^{n-1} \frac{1}{|a - w_j|^\lambda} \geq (n-1) \left(\frac{2}{D} \right)^\lambda \geq n - 1.$$

□

The same argument gives a sharp nearest-critical-point consequence, now normalized by the actual zero diameter rather than by an ambient disk.

Corollary 4.3. *Under the hypotheses of Theorem 4.1, every zero a of p has a critical point w satisfying*

$$|a - w| \leq \frac{D}{2}.$$

The constant $1/2$ is best possible.

Proof. Taking $\lambda = 1$ in Theorem 4.1, at least one of the $n - 1$ summands is at least $2/D$. The degree-two polynomial with two distinct zeros shows sharpness. $\square$

5. EQUALITY AND RIGIDITY

The proof of Theorem 4.1 contains enough strictness to determine all equality cases. We first treat the reciprocal first moment, where nontrivial equality is possible in every degree.

Proposition 5.1. *Under the hypotheses of Theorem 4.1, equality in (4.2) holds if and only if, after a rigid motion and multiplication by a nonzero constant,*

$$p(z) = (z - \beta)(z - \alpha)^{n-1}, \quad a = \beta,$$

or the reflected configuration with $a = \alpha$. Here $\beta - \alpha = D$.

Proof. The displayed polynomials have critical points α with multiplicity $n - 2$ and

$$\frac{(n-1)\beta + \alpha}{n}$$

with multiplicity one. Their reciprocal first moment at $a = \beta$ is therefore

$$\frac{n-2}{D} + \frac{n}{D} = \frac{2(n-1)}{D},$$

so equality is attained.

Conversely, suppose equality holds and $a = x_k$ is simple. If zeros occur on both sides of a , then Rolle's theorem gives a critical point strictly inside each of the adjacent intervals (x_{k-1}, a) and (a, x_{k+1}) . In the proof of Proposition 3.1, this makes both inequalities $Q_- \geq P_-$ and $Q_+ \geq P_+$ strict. Hence (3.2), and therefore (4.2), is strict. Equality thus forces a to be an endpoint of the zero interval.

Assume, for definiteness, that $a = \beta$. Then $P_+ = Q_+ = 0$, and the identity in (3.1) gives

$$\sum_{j=1}^{n-1} \frac{1}{|a - y_j|} = Q_- = 2P_-.$$

Every other zero has distance at most D from a , so $P_- \geq (n-1)/D$. Equality is possible only when every one of those distances equals D , that is, when all remaining zeros coincide at α . The case $a = \alpha$ is symmetric. $\square$

The preceding rigidity statement combines with the strict convexity of the power mean to settle every exponent.

Corollary 5.2. *For $\lambda = 1$, equality in (4.1) is characterized by Proposition 5.1. For $\lambda > 1$, equality in (4.1) holds if and only if $n = 2$.*

Proof. When $\lambda > 1$, equality in the power-mean step requires all numbers $|a - w_j|^{-1}$ to be equal. In the equality configuration of Proposition 5.1, these numbers are

$$\frac{1}{D} \quad \text{with multiplicity } n-2, \quad \frac{n}{D} \quad \text{with multiplicity } 1.$$

They are equal exactly when $n = 2$. Conversely, for a quadratic polynomial the unique critical point is the midpoint of the two zeros, so equality holds for every $\lambda \geq 1$. $\square$

We can now state the sharp equality classification in the unit-disk normalization of Corollary 4.2.

Corollary 5.3. *Suppose the zeros of p are collinear and lie in the closed unit disk.*

- (i) *For $\lambda = 1$, equality in (4.3) holds if and only if, for some $|\zeta| = 1$ and $c \neq 0$,*

$$p(z) = c(z - \zeta)(z + \zeta)^{n-1}$$

and the distinguished zero is $a = \zeta$, or in the reflected configuration.

- (ii) *For $\lambda > 1$, equality in (4.3) holds if and only if $n = 2$ and the two zeros are antipodal points of the unit circle.*

Proof. Equality in Corollary 4.2 requires equality in Theorem 4.1 and $D = 2$. Two points of the closed unit disk are at distance two precisely when they are antipodal boundary points. The conclusion follows from Proposition 5.1 and Corollary 5.2. $\square$

6. ANTIDERIVATIVE COORDINATES

The argument above is self-contained, but it is useful to place it in the reciprocal-coordinate framework that appears in recent work on Sendov's conjecture. The next identity is the antiderivative algebra used by Mazur [Maz26, pp. 3–4]; its first-degree coefficient is also the classical logarithmic-derivative identity associated with Meir and Sharma [MS69, pp. 459–467].

Proposition 6.1. *Let*

$$p(z) = c(z - a) \prod_{i=1}^m (z - z_i), \quad m = n - 1,$$

where a is a simple zero, and let $w_1, \dots, w_m$ be the critical points. Define

$$r_i = \frac{1}{a - z_i}, \quad q_j = \frac{1}{a - w_j}.$$

Then

$$\prod_{j=1}^m (1 + q_j z) = \frac{d}{dz} \left[z \prod_{i=1}^m (1 + r_i z) \right]. \quad (6.1)$$

Equivalently, for $0 \leq k \leq m$,

$$e_k(q_1, \dots, q_m) = (k + 1)e_k(r_1, \dots, r_m),$$

where e_k is the k th elementary symmetric polynomial.

Proof. Direct factorization at a gives

$$\frac{p(a + z)}{p'(a)} = z \prod_{i=1}^m \left(1 + \frac{z}{a - z_i} \right) = z \prod_{i=1}^m (1 + r_i z).$$

Differentiating and factoring p' at its zeros yields

$$\frac{p'(a + z)}{p'(a)} = \prod_{j=1}^m \left(1 + \frac{z}{a - w_j} \right) = \prod_{j=1}^m (1 + q_j z),$$

which proves (6.1). Coefficient comparison gives the symmetric-function identities. $\square$

For $k = 1$, Proposition 6.1 gives

$$\sum_{j=1}^m q_j = 2 \sum_{i=1}^m r_i.$$

For general complex configurations, this identity controls a signed complex sum and does not by itself control the sum of moduli. In the collinear case, however, a rigid motion makes every q_j and r_i real. Proposition 3.1 is precisely the additional order information that separates their positive and negative parts. Thus the collinear proof may be viewed as the antiderivative identity supplemented by the strongest possible one-dimensional phase information: the phase set has collapsed to $\{0, \pi\}$, and Rolle interlacing determines how the two signs are distributed.

This observation also explains why the diameter-normalized inequality is stronger than the ordinary line-case form of Sendov's conjecture. The algebraic identity supplies a factor of two, while the zero counts on the two sides of a convert the larger signed reciprocal mass into the geometric scale $(n - 1)/D$.

7. CONCLUDING REMARKS

Theorem 4.1 gives a local negative-moment inequality at every zero, whereas classical majorization and Schoenberg-type inequalities usually compare global collections of zeros and critical points. The sharp constant depends only on the diameter of the zero set and not on the degree-dependent placement of the distinguished zero. In particular, the result simultaneously contains the ordinary Sendov conclusion, the stronger nearest-critical-point estimate $D/2$, and the entire Tang–Zhang range $\lambda \geq 1$ for collinear zeros.

The proof also indicates what is lost outside the collinear setting. The coefficient identity (6.1) remains valid, but the reciprocal coordinates acquire arbitrary complex phases and there is no Rolle ordering with which to separate the two signed masses. Any extension of the present argument to noncollinear configurations would therefore require a genuinely two-dimensional substitute for the order structure used in Proposition 3.1.

REFERENCES

- [BRS85] B. D. Bojanov, Q. I. Rahman, and J. Szynal, *On a conjecture of Sendov about the critical points of a polynomial*, Math. Z. **190** (1985), no. 2, 281–285. doi:[10.1007/BF01160464](https://doi.org/10.1007/BF01160464)
- [Bor98] J. Borcea, *Two approaches to Sendov's conjecture*, Arch. Math. (Basel) **71** (1998), no. 1, 46–54. doi:[10.1007/s000130050232](https://doi.org/10.1007/s000130050232)
- [Bra68] D. A. Brannan, *On a conjecture of Ilieff*, Proc. Cambridge Philos. Soc. **64** (1968), 83–85. doi:[10.1017/S0305004100042596](https://doi.org/10.1017/S0305004100042596)
- [BP07] J. E. Brown and V. F. Powell, *A result on real polynomials with real critical points*, J. Anal. Appl. **5** (2007), no. 1, 41–52.
- [BX99] J. E. Brown and G. Xiang, *Proof of the Sendov conjecture for polynomials of degree at most eight*, J. Math. Anal. Appl. **232** (1999), no. 2, 272–292. doi:[10.1006/jmaa.1999.6267](https://doi.org/10.1006/jmaa.1999.6267)
- [Cha20] T. P. Chalebgwa, *Sendov's conjecture: a note on a paper of Dégot*, Anal. Math. **46** (2020), no. 3, 447–463. doi:[10.1007/s10476-020-0050-x](https://doi.org/10.1007/s10476-020-0050-x)
- [Deg14] J. Dégot, *Sendov conjecture for high degree polynomials*, Proc. Amer. Math. Soc. **142** (2014), no. 4, 1337–1349. doi:[10.1090/S0002-9939-2014-11888-0](https://doi.org/10.1090/S0002-9939-2014-11888-0)
- [Mar66] M. Marden, *Geometry of polynomials*, 2nd ed., Mathematical Surveys, vol. 3, American Mathematical Society, Providence, RI, 1966. doi:[10.1090/surv/003](https://doi.org/10.1090/surv/003)
- [Mar83] M. Marden, *Conjectures on the critical points of a polynomial*, Amer. Math. Monthly **90** (1983), no. 4, 267–276. doi:[10.2307/2975758](https://doi.org/10.2307/2975758)
- [Maz26] L. Mazur, *A computer-assisted proof of Sendov's conjecture*, preprint, 2026, 19 pp.
- [MS69] A. Meir and A. Sharma, *On Ilieff's conjecture*, Pacific J. Math. **31** (1969), no. 2, 459–467. doi:[10.2140/pjm.1969.31.459](https://doi.org/10.2140/pjm.1969.31.459)
- [Mil90] M. J. Miller, *Maximal polynomials and the Ilieff–Sendov conjecture*, Trans. Amer. Math. Soc. **321** (1990), no. 1, 285–303. doi:[10.1090/S0002-9947-1990-0965744-X](https://doi.org/10.1090/S0002-9947-1990-0965744-X)

- [Mil08] M. J. Miller, *Unexpected local extrema for the Sendov conjecture*, J. Math. Anal. Appl. **348** (2008), no. 1, 461–468. doi:[10.1016/j.jmaa.2008.07.049](https://doi.org/10.1016/j.jmaa.2008.07.049)
- [Mil25] M. J. Miller, *Seeking a quadratic refinement of Sendov’s conjecture*, preprint, 2025, arXiv:2506.12951v2, 10 pp.
- [PR72] D. Phelps and R. S. Rodriguez, *Some properties of extremal polynomials for the Ilieff conjecture*, Kōdai Math. Sem. Rep. **24** (1972), 172–175. doi:[10.2996/kmj/1138846519](https://doi.org/10.2996/kmj/1138846519)
- [RS02] Q. I. Rahman and G. Schmeisser, *Analytic theory of polynomials*, London Mathematical Society Monographs. New Series, vol. 26, The Clarendon Press, Oxford University Press, Oxford, 2002. doi:[10.1093/oso/9780198534938.001.0001](https://doi.org/10.1093/oso/9780198534938.001.0001)
- [Rub68] Z. Rubinstein, *On a problem of Ilieff*, Pacific J. Math. **26** (1968), no. 1, 159–161. doi:[10.2140/pjm.1968.26.159](https://doi.org/10.2140/pjm.1968.26.159)
- [ST71] E. B. Saff and J. B. Twomey, *A note on the location of critical points of polynomials*, Proc. Amer. Math. Soc. **27** (1971), no. 2, 303–308. doi:[10.1090/S0002-9939-1971-0271312-1](https://doi.org/10.1090/S0002-9939-1971-0271312-1)
- [Sch69] G. Schmeisser, *Bemerkungen zu einer Vermutung von Ilieff*, Math. Z. **111** (1969), 121–125. doi:[10.1007/BF01111192](https://doi.org/10.1007/BF01111192)
- [Sch71] G. Schmeißer, *Zur Lage der kritischen Punkte eines Polynoms*, Rend. Sem. Mat. Univ. Padova **46** (1971), 405–415.
- [Sch04] G. Schmeisser, *Majorization of the critical points of a polynomial by its zeros*, Comput. Methods Funct. Theory **3** (2004), no. 1, 95–103. doi:[10.1007/BF03321027](https://doi.org/10.1007/BF03321027)
- [Sch86] I. J. Schoenberg, *A conjectured analogue of Rolle’s theorem for polynomials with real or complex coefficients*, Amer. Math. Monthly **93** (1986), no. 1, 8–13. doi:[10.2307/2322536](https://doi.org/10.2307/2322536)
- [SAG20] G. M. Sofi, S. A. Ahanger, and R. B. Gardner, *Some classes of polynomials satisfying Sendov’s conjecture*, Studia Sci. Math. Hungar. **57** (2020), no. 4, 436–443. doi:[10.1556/012.2020.57.4.1475](https://doi.org/10.1556/012.2020.57.4.1475)
- [SS23] G. M. Sofi and W. M. Shah, *On Sendov’s conjecture*, Rend. Circ. Mat. Palermo (2) **72** (2023), no. 1, 493–497. doi:[10.1007/s12215-021-00690-y](https://doi.org/10.1007/s12215-021-00690-y)
- [Tao22] T. Tao, *Sendov’s conjecture for sufficiently high degree polynomials*, Acta Math. **229** (2022), no. 2, 347–392. doi:[10.4310/ACTA.2022.v229.n2.a3](https://doi.org/10.4310/ACTA.2022.v229.n2.a3)
- [TZ25] Q. Tang and T. Zhang, *Sharp Schoenberg type inequalities and the de Bruin–Sharma problem*, preprint, 2025, arXiv:2508.10341v3, 57 pp.
- [Tot21] V. Totik, *Critical points of polynomials*, Acta Math. Hungar. **164** (2021), 499–517. doi:[10.1007/s10474-021-01133-x](https://doi.org/10.1007/s10474-021-01133-x)
- [Zha26] T. Zhang, *Sendov’s conjecture holds for every degree $n \geq 10^{200000}$* , preprint, 2026, 29 pp., https://zhangteng2000.github.io/files/sendov_conjecture_large_n.pdf.

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